

LEAF: A Robust Multimodal Federated Learning Framework for Tea Pest and Disease Diagnosis

Ayush Debnath

Ruelia Saha

Sudip Misra

Damodhara R. Mailapalli

Abstract—Tea diagnosis combines lesion imagery and field notes, but small corpora can inflate multimodal gains through source overlap and label-bearing text. We present LEAF (Leakage-audited Evidence-Aware Fusion), a reliability-routed multimodal federated framework evaluated on 371 annotated crops from 200 tea photographs. Exact box-note linkage, source-disjoint 222/74/75 splits, and a training-fitted short-cut mask control leakage. Validation retains the ResNet-Transformer router over a ViT-tiny-DistilBERT candidate; their 61.33% and 64.00% test accuracies differ by only two crops (McNemar $p = 0.82$). Separately, frozen ViT-Base with BERT-small reaches 80.00% on the same support, exceeding image-only (73.33%) and text-only (64.00%) baselines. Full modality ablations change 41/75 predictions for text and 49/75 for zeroed images, although mismatched notes cost only 2.67 accuracy points. FedAvg retains 84–94% of centralized validation macro F1 across 2–8 simulated clients. Federation improves on local-only training at 18.9–67.2 MiB per client-round; fusion tolerates tested skew and dropout but degrades under stale and poisoned updates. FarmFederate demonstrates photo/note capture, diagnostic feedback, and advisory chat. Templated notes, rare classes, simulated clients, and prototype telemetry limit deployment claims; multi-estate validation remains necessary.

Index Terms—Federated Learning, Vision-Language Models, Tea Pest and Disease Classification, Multimodal Learning, Evaluation Protocol, Data Leakage, Precision Agriculture

I. Introduction

TEA (*Camellia sinensis*) suffers visually confusable pests and diseases that reduce yield and leaf quality [19]. Photographs capture visible damage, while scouts record symptom progression, weather, and pest pressure. Lightweight image models support field recognition [1], but centralizing estate records raises connectivity and data-ownership concerns [20]. Federated learning (FL) offers collaborative training through model updates [6].

Reliable evaluation is equally important. Crops from one photograph can leak background context across random splits; disease names and class-pure phrases can reveal targets directly. LEAF therefore groups source photographs, links notes to exact boxes, and fits text filtering on training data alone. Our contributions are (i) a traceable five-class crop-note corpus and leakage audit; (ii) common-support model, modality, and corruption comparisons; (iii) validation-selected routing and three-seed federated adaptation with cost and failure-mode analysis; and (iv) a companion mobile workflow connecting observation entry to readable results and follow-up advice.

II. Related Work

Recognition and federation. Lightweight tea diagnosis [1] and broader AI disease-management work [2] motivate field deployment. Agricultural FL extends FedAvg with explainability and hierarchical sensing [4], [5]. Our evaluation adds source-disjoint image-note linkage, realized client skew, communication cost, local-only controls, and matched aggregator comparisons.

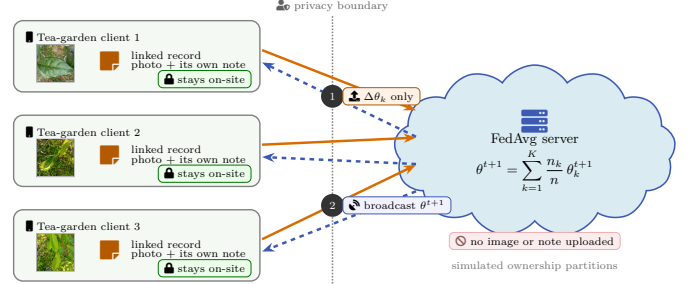


Fig. 1: Federated workflow. Simulated clients retain photos and notes; model updates go to FedAvg and global parameters return.

Multimodal learning. Contrastive image-text alignment [8] and frozen-encoder fusion [9] inform multimodal crop recognition [10], feature-replay federation [11], and sensor-image fusion [18]. We compare compact text encoders [16] and visual backbones [3], [7], [12]–[15] on fixed support. Class-generated or class-matched notes can partially encode the target; our templated notes remain a limitation despite exact linkage and training-only shortcut filtering.

III. System Model

Each estate keeps images and notes local; only model updates cross the boundary (Fig. 8). The controlled ablations use one selected checkpoint, while FL adapts its trainable multimodal states with the visual backbone frozen. Field-note words are BLAKE2b-hashed into a fixed 30522-entry ID space and padded or truncated to 128 tokens; ImageNet-normalized 224×224 images become a 7×7 grid of 49 ResNet-50 spatial tokens for bidirectional cross-attention.

A. Dataset Procurement, Preparation, and Leakage Audit

Held fixed. Every table except Table II is scored on the same 222/74/75 source-grouped crops, the same exact box-note pairing and the same train-fitted leakage mask, so any difference between them is the axis that table varies. The matched-setting federated suite of Table II re-partitions those same 371 observations as 260/55/56 over 199 source groups, and is read only against itself.

Procurement and preparation. The corpus contains 200 field photographs from the IIT Kharagpur tea garden, supplied as JPEGs with same-stem YOLO oriented-bounding-box (OBB) files. Each valid line records a class ID and eight normalized corner coordinates. The loader retains each region’s supplied label, extracts its axis-aligned envelope, adds 10% of the source width/height as context on each side, clips to image bounds, and resizes to 224×224 . This yields 371 crops: leaf blight (129), leaf hoppers (9), leaf rust (67), looper caterpillars (102), and mosquito bug (64), using raw IDs 0–4. The

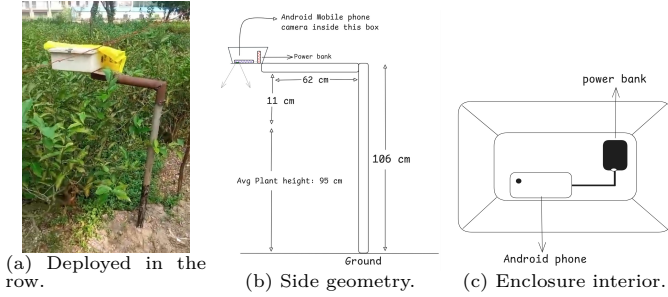


Fig. 2: Capture rig used to collect the corpus. (a) The staked pole and arm in the tea row. (b) A 106 cm pole with a 62 cm arm places the camera 11 cm above the 95 cm mean canopy. (c) An Android handset and power bank inside the weather enclosure. The fixed geometry keeps scale and viewpoint comparable across photographs; no capture time is logged.

TABLE I: Source-grouped paired-data audit

Split	Sources	Crops	Blight	Hoppers	Rust	Looper/Mosq.
Train	122	222	77	5	40	61/39
Validation	40	74	26	2	14	20/12
Test	38	75	26	2	13	21/13
Total	200	371	129	9	67	102/64

acquisition rig is documented in Figure 2, but capture dates, the handset model and sensor specification, and independent expert adjudication are not recorded in the supplied bundle. Figure 3 shows one real annotated crop per class; its manifest preserves source filenames, box indices, and crop bounds. The supplied labels are not independent confirmatory diagnoses.

Acquisition rig. Photographs were captured by a fixed, self-contained station rather than handheld, which is what makes the corpus internally consistent in scale and viewpoint. An Android handset is sealed in a weather enclosure with a power bank (Figure 2c) and clamped to a horizontal arm on a ground-staked pole (Figure 2a). The pole stands 106 cm, the arm extends 62 cm over the row, and the camera sits 11 cm above a mean canopy height of 95 cm, giving a consistent nadir view of the plucking table (Figure 2b). Because the geometry is fixed, apparent lesion size is comparable across photographs, and the 10% context padding applied to each box corresponds to a roughly constant ground area. The enclosure is unpowered from mains and needs no network, so the same rig transfers to an estate without infrastructure — the deployment assumption the federated setting is meant to model. The rig does not record capture time, which is why no date is available for any crop.

Split and text provenance. Seed 42 partitions source filenames, never crops: overlap is zero, and test holds 75 crops from 38 unseen photographs (Table I). Every note links to its exact (photograph, box index). Notes are generated from manually written phrase pools, not recorded independently by farmers: 140 distinct sentences supply 371 notes, 61% of sentences occur across classes, and a class’s diagnostic sign appears in 45% of its notes. A bag-of-words classifier reaches 0.640. The 178-token training-derived target-pure vocabulary is frozen before evaluation. The separate 3,000-row text bundle, augmented exports, and external studio collections are excluded; ImageNet supplies weights only. The nine-crop leaf-hopper class and templated language limit generalization.

B. Model Variants and Objective

The reliability network is a two-layer MLP over the concatenated embeddings whose two softmax-normalized outputs give weights $r_t + r_v = 1$ — how far the fused prediction leans on the note and on the image. An absent modality’s logit is set to -10^4 before the softmax, so its weight goes to zero and the survivor takes the full unit of mass. The final logits are

$$\mathbf{z} = \mathbf{z}_f + \lambda (r_t \mathbf{z}_t + r_v \mathbf{z}_v), \quad \lambda = 2, \quad (1)$$

where $\mathbf{z}_f$ is the fusion head’s five-way output and $\mathbf{z}_t, \mathbf{z}_v$ come from two linear expert heads reading each embedding directly. The residual weight $\lambda = 2$ is fixed rather than learned, so the two single-modality experts count twice the fused head and a usable prediction survives when one branch is absent; a missing modality’s expert is zeroed and the residual stays active at test time.

Comparison models. Frozen DistilBERT [16] and ViT-tiny [7] use validation-selected ridge heads. Their late-fusion candidate uses train-fitted 64-component projections and one of 360 validation-screened linear heads; test embeddings are extracted only after the choice is frozen. The raw and routed VLM rows are the cross-attention output before and after the reliability router.

For one-hot label $\mathbf{y}$, training uses class-weighted, label-smoothed cross entropy — weights inverse to class frequency, smoothing 0.1 — with auxiliary parent losses and supervised cross-modal alignment:

$$\mathcal{L} = \mathcal{L}_f + 0.20\mathcal{L}_t + 1.25\mathcal{L}_v + 0.05\mathcal{L}_{align}. \quad (2)$$

Here $\mathcal{L}_f$ is the loss on the fused logits; $\mathcal{L}_t$ and $\mathcal{L}_v$ apply it to the two expert heads, averaged only over samples where that modality is present, so a text-only batch never trains the image expert; $\mathcal{L}_{align}$ is a supervised contrastive term over the ℓ_2 -normalized paired embeddings, evaluated only when a batch holds at least two paired crops. The three weights are fixed by construction rather than tuned: image-only learning is the harder path while templated notes make the text expert easy to satisfy, and alignment is a regularizer that must not compete with the classification objective. Modality dropout samples image-only, text-only and paired batches with probabilities 0.50, 0.10, 0.40; checkpoints are selected by $0.65F_{1,paired} + 0.35F_{1,image}$ on validation.

For K clients, the same local objective is aggregated with FedAvg [6],

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}, \quad (3)$$

where n_k is client k ’s sample count. FL starts from the selected checkpoint, freezes ResNet-50 and averages the remaining states after one local epoch: adaptation, not training from scratch.

IV. Evaluated Systems

A. Base Cross-Modal Checkpoint

The checkpoint trains in PyTorch for at most 15 AdamW epochs (batch 16, peak 10^{-4}). Two cached augmentations per training crop remain views of 222 crops, not new samples.

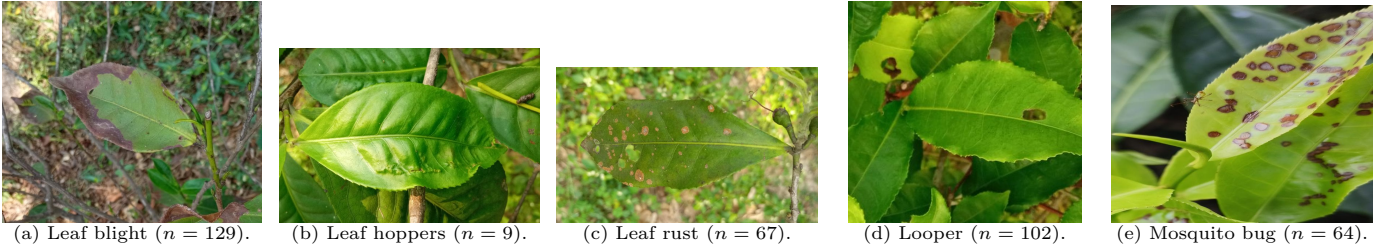


Fig. 3: One actual annotated crop per class; n is the corpus crop count. Source OBB labels and 10% context padding are preserved. Natural backgrounds and overlapping damage patterns complicate recognition; no external or generated images are shown.

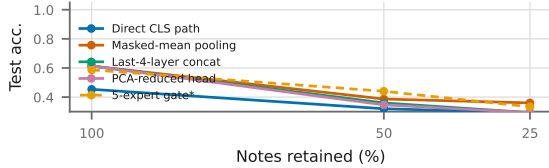


Fig. 4: Text-only accuracy on 75 crops under note deletion; the best rung falls from 0.613 to 0.360. Dashed gates (*) are diagnostic.

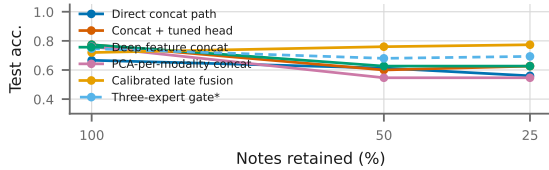


Fig. 5: Fusion under the same deletion: the best rung retains at least 0.760 while text-only falls to 0.360.

B. Enhanced Image-Only Inference

Five single deployable visual systems compete on the common support, ranked by the pre-declared validation score $0.6F_1 + 0.4$ accuracy, with the locked test support reported after the choice is frozen. Across the eight visual rungs, locked test accuracy rises from 0.613 to 0.720 (Fig. 6); Figs. 4 and 5 are the corresponding text and text+image ladders. Cand. counts the configurations each rung searched, and the gate rungs route among several encoders, which is why they are marked as diagnostics.

The same ladder runs for text and text+image on the identical support, so every rung is comparable with the others and with the visual ladder. Neither ladder is flat now that the notes no longer determine the label. The best text rung reaches 0.613 at complete notes and 0.360 once three quarters of the tokens are gone, while the best fusion rung holds 0.773 at both (Fig. 5): the fused pipeline is what survives note loss, and the validation-to-test gap still widens with search (Fig. 7).

The calibrated base wins on both sides of the split on the smallest search in the table (validation 0.771, test accuracy 0.680); the gates score higher only by fitting more routings on 74 crops, and Fig. 7 shows what that costs. We carry the calibrated base as the visual path and report the gate as a diagnostic ceiling.

C. Reliability-Aware Sparse-Note Rule

For sparse notes, a stable hash retains each non-special token with probability 0.50. A validation-fixed rule routes low-confidence (< 0.60), base-predicted leaf-rust cases to

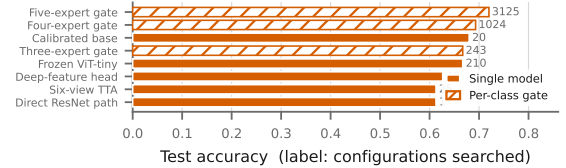


Fig. 6: Alternative image-only systems and search budgets. Accuracy rises from 0.613 to 0.720; hatched class-specific gates are diagnostic, not deployable models.

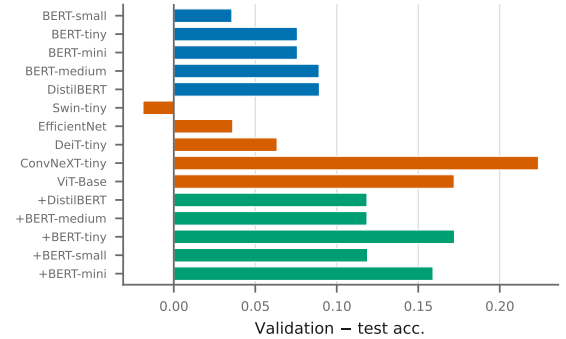


Fig. 7: Validation minus test accuracy per system on the aligned support. A large positive gap is selection that did not transfer from 74 validation crops.

the sparse-text expert; otherwise it keeps late fusion. Because the routing family was refined on this internal partition, its test score is exploratory and non-pristine.

D. Multimodal Federated Adaptation

FL adapts 8.69M parameters with frozen ResNet-50, one local epoch, full participation, and sample weighting. We sweep $K = \{2, 3, 5, 8\}$ for eight rounds over seeds 42, 123, and 206 on the 74-crop validation set. Clients are simulated; the ensemble and router are excluded from this FL score.

The two halves of a round separate cleanly: every stage touching a photograph, box, or note runs on the client.

E. Advisory Module

The Classify–Retrieve–Advise module embeds a query with Sentence-BERT [17] and performs exact FAISS inner-product search. Its corpus is textual only and shares no records with the box-linked crop set: 673 unique observations (132–138 per class) built from 131 distinct templated sentences. A deduplicated split leaves 472 passages and 201 queries with no shared observation. Correctness requires a class match.

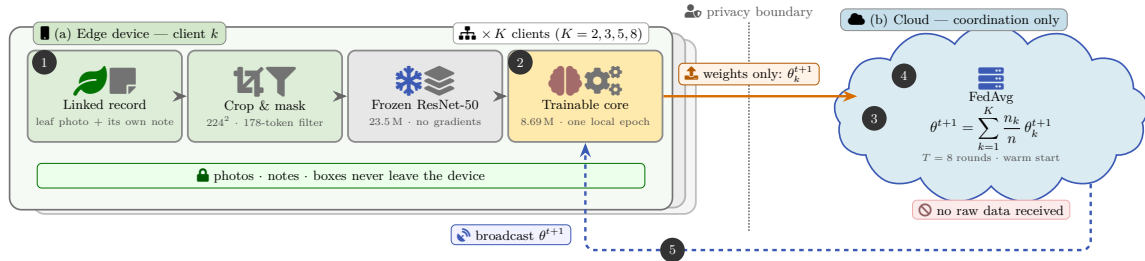


Fig. 8: One FedAvg round: box-linked loading, cropping, masking, and local training of 8.69M parameters remain on the client. The server averages updates; fusion uplink costs 59.5–60.5 MiB per client-round.

V. FarmFederate Application and Practical Use

FarmFederate is the Flutter advisory prototype associated with LEAF. Figure 9 shows its six screens. Sign-in leads to a dashboard (a) with crop-check/chat shortcuts, service status, and sensor cards. “Check My Crops” (b) accepts a camera or gallery photograph, a plain-language symptom description, or both; example prompts and optional sensor context support entry, and analysis settings expose text, image, and fusion choices. Results (c) present a status, category scores, and recommended actions, while AI chat (d) supports follow-up questions with text and image input. Analytics (e) and farm-network (f) screens present trend and round/participant views.

Practical value. A scout can photograph a suspect leaf, describe its progression, inspect feedback, and ask for clarification in one interface, which organizes field evidence into a shared starting point for extension-worker review; readable results and conversation help growers interpret model outputs, and monitoring views give supervisors context. These are intended workflow benefits: time savings, adoption, yield gains, and reduced input use were not measured.

Prototype scope. Screenshot scores and broad stress labels such as “Needs Water” are demonstration content, distinct from the five-class tea results. Analytics history is simulated and farm-network state is preset, so these screens do not establish live telemetry or multi-estate FL; sensor input is not a third evaluated modality. The client sends images and notes to a configured inference backend, so the FL training boundary does not imply on-device inference. Deployment requires the selected tea checkpoint, its exact label schema, measured services, and field validation.

VI. Performance Evaluation

A. Protocol

All crops from a source photograph remain in one partition. Model selection and the five-expert visual gate use only the 74-crop validation partition; every number below uses the same 75-crop, 38-source internal test support. Macro F1 accompanies accuracy because the classes are imbalanced. Source-group bootstrap intervals (5,000 replicates) resample source photographs rather than crops.

Five evidence tracks remain separate: the exact-pair same-checkpoint ablation; the validation-selected family comparison on the common 75-crop support; the 74-crop validation-only FedAvg sweep; earlier text, vision, and proxy-fusion screens whose differing supports permit only within-panel ranking; and the 371-pair matched-setting suite on genuine box-linked observations, with no label-matched resampling and no synthetic images.

B. Federated Client and Round Scaling

Figs. 10 and 11 answer the two federated scaling questions directly. Final-round macro F1 falls from 0.6527 at $K = 2$ to 0.5846 at $K = 8$ against a centralized initialization of 0.6919, so 84–94% is retained: adaptation costs something on the corrected corpus where it cost almost nothing on the saturated one. Round-wise means stay between 0.5735 and 0.6857, the highest a $K = 3$ transient at round 7 rather than the final model. Realized label-distribution total variation rises with K from 0.151 to 0.281 without a monotone final-score penalty, over 3 partition seeds.

This result is narrow: every run begins from the same centrally selected checkpoint, so it supports federated adaptation of the multimodal core, not superiority over training from scratch. Three partition seeds measure ownership-split variability, not checkpoint-training variance, and simulated clients cannot establish estate-level privacy.

C. Matched-Setting Controlled Ablations

Table II closes the remaining gaps without mixing supports. The note correction rewrote text only, so the corrected splitter still realizes label TV 0.451 against 0.265 at $\alpha = 0.1$. Federation beats local-only training for all five architectures (+0.124 to +0.229), and the ordering is architectural rather than the note-leakage ceiling the previous corpus imposed. No aggregator dominates — FedProx leads on both fusion variants, FedAvg on image-only and cross-attention, FedBN on text — while SCAFFOLD is unstable everywhere (0.130–0.294). Communication spans 3.5×, 18.9 to 67.2 MiB per client-round, so image-only reaches 0.513 macro F1 at roughly a third the uplink of either fusion (Fig. 12). Two arms report nothing, shown rather than asserted: the anti-collapse components move the mean by at most 0.044 (Fig. 13), and warm start leads on two of three seeds but loses the mean by 0.006, with the fusion variants inside the same seed spread (Fig. 14). Neither is claimed necessary.

Table III extends the federated evidence beyond one architecture. A closed-form ridge head has no gradients for FedAvg to average, so these systems are federated as linear probes — frozen encoder, SGD head — which is not the centralized estimator. Within this protocol fusion leads at both skews (0.723 against 0.607 for text and 0.508 for image at $\alpha = 0.1$; 0.775, 0.628 and 0.559 at $\alpha = 1$) and takes the largest federation gain over local-only training in both cells (+0.406 and +0.411), and it holds at every round (Fig. 15) and every client count: at $\alpha = 1$ fusion spans 0.734–0.799 from $K = 2$ to 50, above text (0.587–0.689) throughout (Fig. 16). Combining the

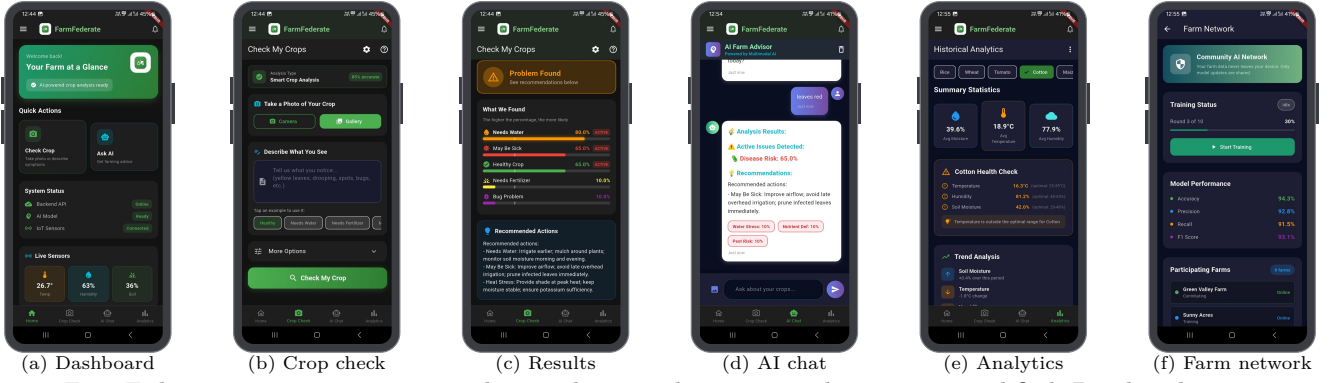


Fig. 9: FarmFederate prototype screens in drawn phone outlines; screenshots are unmodified. Displayed scores, sensor values, and broad stress labels are demonstration content, and (e)–(f) show simulated history and preset network state, not audited LEAF results.

TABLE II: Matched federated suite: 371 box-linked crops, 260/55/56 split over 199 source groups, no synthetic images. Architectures share client partitions; entries are three-seed validation macro-F1 means over 50 rounds.

(a) FedAvg by architecture and skew					(b) Aggregator at $\alpha=0.1$					(c) Cost and federation gain		
Architecture	$\alpha=.1$	$\alpha=.5$	$\alpha=1$	$\alpha=10$	Arch.	Avg	Prox	SCAF	BN	Architecture	MiB	Δ local
Image only	0.513	0.516	0.479	0.486	Image	.513	.504	.222	.196	Image only	18.9	+0.215
Text only	0.468	0.448	0.441	0.431	Text	.468	.467	.220	.470	Text only	42.4	+0.124
Concat VLM	0.531	0.551	0.548	0.546	Concat	.531	.560	.179	.427	Concat VLM	59.5	+0.229
Attention VLM	0.541	0.555	0.524	0.561	Attention	.541	.549	.130	.427	Attention VLM	60.5	+0.219
Cross-attn. VLM	0.499	0.485	0.466	0.512	Cross-att	.499	.482	.294	.455	Cross-attn. VLM	67.2	+0.138

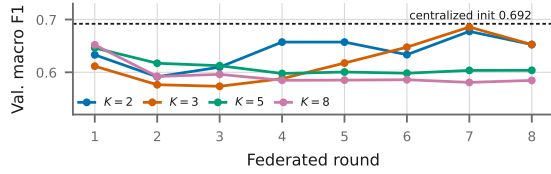


Fig. 10: Rounds against performance for the retained core, one line per client count. Dashed: the centralized initialization each run adapts from. $K=2$ and 3 climb toward it; $K=5$ and 8 are flat.

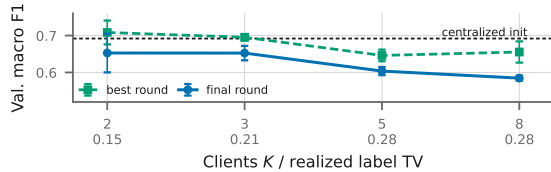


Fig. 11: Clients against performance for the retained core: the same fixed training set split over more owners. Bars are ± 1 SD over 3 seeds; realized label TV is printed under each tick.

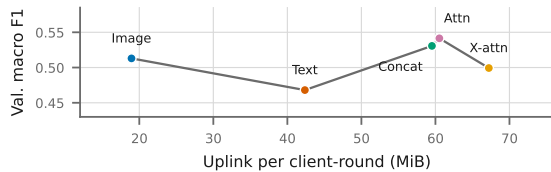


Fig. 12: Uplink cost against accuracy at $\alpha=0.1$; the line orders architectures by uplink rather than interpolating. Cost spans $3.5\times$, accuracy 0.073.

branches therefore helps most exactly where partitioning is most severe. Client partitions remain simulated.

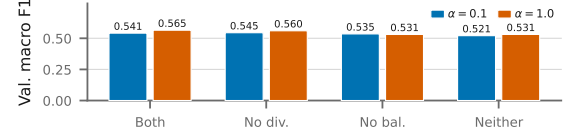


Fig. 13: Anti-collapse arms at both skews, from zero because the finding is that none of them collapses. Values are printed: the arms span 0.044, all of it the balanced sampler, and the diversity term does nothing.

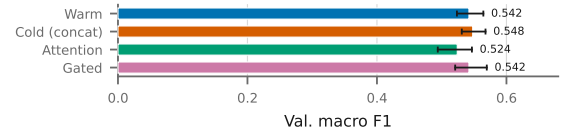


Fig. 14: Every single-architecture matched-setting arm: bars are 3-seed means, whiskers the observed seed range. The cold and concat arms are the same runs. The ranges overlap, so no arm clears the spread.

TABLE III: Frozen-encoder SGD heads on the shared source-grouped split. Each cell averages 18 runs (three seeds, six client counts; 50 rounds). Local denotes no communication; client partitions are matched within each cell.

System	$\alpha = 0.1$		$\alpha = 1$	
	FedAvg	Local	FedAvg	Local
Text (frozen DistilBERT)	0.607	0.266	0.628	0.298
Image (frozen ViT-tiny)	0.508	0.323	0.559	0.391
ViT+DistilBERT fusion	0.723	0.317	0.775	0.364
ResNet+compact (warm)	0.892–0.910 over 2–8 clients			

D. Same-Checkpoint Ablation and Image-Only Gains

Both branches carry evidence and neither dominates: unimodally the image leads this test 0.6400 against 0.2400, and Section VI-H quantifies both ablations. Notes are

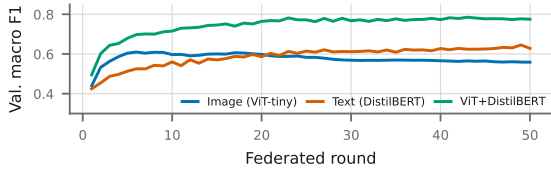


Fig. 15: Rounds against performance for the frozen-encoder systems at $\alpha=1$ (three seeds $\times$ six client counts). Fusion leads at every round: 0.775 against 0.628 text and 0.559 image.

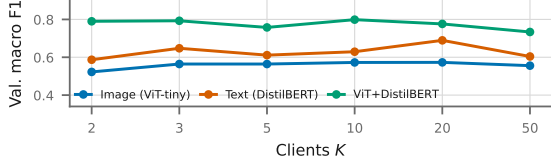


Fig. 16: The same systems against client count. Fusion leads at every count, 0.734–0.799 against 0.587–0.689 for text.

templated, so every note-side number is an upper bound, and the routing family was refined on this partition. A validation-selected test-time-augmentation search over 25 view/scale candidates reaches 74.32% validation accuracy (macro F1 0.6854) with the base, tight-crop and context-crop views, and 64.00% on the locked test (macro F1 0.5532). This is an inference-time ensemble over one backbone, not a stronger encoder; rare-class behaviour remains unstable.

E. Unimodal and Multimodal Systems, Centralized and Federated

Tables IV and V place a classical feature baseline, five frozen text encoders, five frozen vision encoders and their fusions on the same 222/74/75 support, scored centrally and again under FedAvg.

Unimodal. With complete notes the image branch leads the text branch: ViT-Base reaches 0.733 against 0.640 for the best text encoder, BERT-small. Federating costs little on either side (0.733 to 0.667; 0.640 to 0.582), so the modality separates these systems, not the deployment mode. The classical rows locate where pretraining earns its place: decisive on images (0.440 against 0.733) and absent on text, where TF-IDF with a linear SVM ties the best transformer at 0.640. Neither branch is near the ceiling, so the comparison is informative rather than saturated.

Multimodal. Fusion beats both parents outright, and does so at complete notes: ViT-Base with BERT-small reaches 0.800 against 0.733 for the best image-only encoder and 0.640 for the best text-only encoder. All five ViT-Base+BERT rows clear the best unimodal system (lowest 0.747 against 0.733), so the gain is not one lucky pairing; the classical TF-IDF with colour/HOG fusion does not, placing the gain in the pretrained encoders rather than in concatenation. The advantage widens as notes degrade: at 75% deletion text falls to 0.373 while fusion holds 0.787 and the image branch is unmoved by construction. Federation preserves the ordering rather than creating it, and its cost falls as the note thins — 0.800 to 0.667 at complete notes, 0.787 to 0.716 at 75% deletion.

Every ViT-Base pairing outperforms the best text-only system at every sparsity level, but the best text partner

changes: BERT-small leads with complete notes, while BERT-mini leads at 75% sparsity. The performance spread across the five pairings widens from 0.053 to 0.187. These results support fusion over its unimodal parents, but do not identify a consistently best text encoder.

F. Robustness: Skew, Dropout, Stragglers and Poisoning

Skew and dropout break nothing (Fig. 17a). From $\alpha=10$ down to $\alpha=0.01$, and under a pathological partition giving each client a single class, fused macro-F1 stays in 0.741–0.776 — its worst setting above the best setting of either parent (0.617 text, 0.660 image). Dropout is flatter still: 70% of clients failing to report each round leaves the fused system at 0.773 against 0.772 with none. The image branch is the limiting case: a convex probe returns per-seed-identical scores across $\alpha=0.5, 1$ and 10 and again across $\alpha=0.05, 0.01$ and the pathological split, so skew changes the path and not the destination.

Stragglers cost what dropout does not (Fig. 17b). A stale update arrives and is averaged in, pulling the model toward an older global state rather than merely shrinking the cohort. At 75% stale updates per round the fused system loses 0.115 (0.772 to 0.657) and the image branch 0.123 (0.660 to 0.537), where the same fraction of dropped clients costs nothing. Discarding a late update beats waiting for it.

Poisoning is where plain FedAvg has no defence (Fig. 17c). Sign-flipped updates are catastrophic at every fraction: 10% malicious clients take the fused system from 0.772 to 0.298 and 30% to 0.123, with text (0.136) and image (0.104) equally destroyed — the one condition in which fusion is not best, because all three are broken. Label flip degrades gradually instead (0.658, 0.611, 0.539 at 10/20/30%) and Gaussian noise sits between them. Coordinate-wise median recovers the outlier attacks (sign flip to 0.613, 0.543, 0.420) but not label flip, which is worse at 30% (0.510 against 0.539): a label-flipped update is a well-formed gradient step toward the wrong target, not an outlier in parameter space. The median also costs 0.057 with nobody attacking, so it is a trade, not a free defence. Nothing here is Byzantine-tolerant.

G. Actual Model-Family and Inter-Model Comparison

In Table VI the routed ResNet-compact model, LEAF, is the incumbent — the core already carried into the routing and federated experiments — and the pooled ViT-DistilBERT model is the candidate, adopted only if it beats the incumbent on validation. Validation selects the incumbent before test extraction: 77.03% accuracy and 0.7169 F1 against 72.97% and 0.5980 for the candidate. On test the candidate predicts 2 more crops, but the 11-versus-9 discordance is not significant (McNemar $p = 0.82$) and the two macro-F1 values differ by 0.0014. On 75 crops the support does not separate them, so the selection rule decides and the test column is reported rather than ranked. The enhanced gate gains eight correct crops over the direct image path and the router 15, mainly in blight and looper; two leaf-hopper crops are too few to carry a claim.

H. Missing-Modality and Corruption Probes

Mismatching the note costs only 2.67 accuracy points and 0.0187 F1, and exact cross-modal recall@1 is 1.33%

TABLE IV: Unimodal systems on the same 222/74/75 split, crop-linked notes, and train-fitted mask. Test metrics use validation-selected configurations. FedAvg uses five Dirichlet($\alpha = 1$) clients, 50 rounds, and three seeds. Centralized heads are closed-form; federated heads use SGD. Image rows are invariant to note deletion. Bold: best per block/column.

System	Dim.	Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
		Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
Text only													
TF-IDF + linear SVM	864	0.640	0.503	0.582	0.469	0.493	0.390	0.476	0.375	0.373	0.298	0.360	0.293
BERT-tiny	128	0.613	0.508	0.440	0.332	0.400	0.316	0.324	0.251	0.320	0.204	0.276	0.196
BERT-mini	256	0.613	0.496	0.427	0.331	0.427	0.318	0.440	0.332	0.280	0.186	0.276	0.190
BERT-small	512	0.640	0.512	0.564	0.467	0.427	0.306	0.431	0.331	0.373	0.291	0.253	0.197
BERT-medium	512	0.600	0.493	0.529	0.426	0.333	0.254	0.400	0.309	0.253	0.158	0.213	0.185
DistilBERT	768	0.613	0.485	0.529	0.422	0.427	0.328	0.431	0.332	0.360	0.286	0.249	0.194
Image only													
Colour+HOG SVM	630	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392	0.440	0.346	0.471	0.392
Swin-tiny	768	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463	0.600	0.507	0.569	0.463
ConvNeXT-tiny	768	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470	0.560	0.493	0.560	0.470
EfficientNet-B0	1280	0.680	0.561	0.680	0.573	0.680	0.561	0.680	0.573	0.680	0.561	0.680	0.573
DeiT-tiny	192	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520	0.693	0.545	0.636	0.520
ViT-Base/16	768	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577

TABLE V: Multimodal systems under the protocol of Table IV. Best unimodal references appear below. Complete-note fusion reaches 0.800 versus 0.733 image-only and 0.640 text-only. Bold: best per column.

System	Dim.	Complete notes				50% of notes deleted				75% of notes deleted			
		Centralized		Federated		Centralized		Federated		Centralized		Federated	
		Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1
Multimodal (image + text)													
TF-IDF + Colour/HOG SVM	1494	0.667	0.537	0.631	0.526	0.587	0.480	0.547	0.451	0.547	0.451	0.449	0.381
ViT-Base + BERT-tiny	896	0.747	0.625	0.693	0.597	0.733	0.614	0.684	0.645	0.600	0.513	0.698	0.610
ViT-Base + BERT-mini	1024	0.773	0.643	0.676	0.585	0.667	0.574	0.684	0.619	0.787	0.655	0.716	0.614
ViT-Base + BERT-small	1280	0.800	0.664	0.667	0.600	0.627	0.543	0.631	0.545	0.613	0.528	0.676	0.572
ViT-Base + BERT-medium	1280	0.773	0.642	0.676	0.566	0.720	0.593	0.653	0.561	0.680	0.571	0.622	0.515
ViT-Base + DistilBERT	1536	0.773	0.645	0.653	0.553	0.693	0.579	0.636	0.554	0.693	0.581	0.640	0.531
Best unimodal system, repeated for reference													
Text only: TF-IDF + linear SVM	864	0.640	0.503	0.582	0.469	0.493	0.390	0.476	0.375	0.373	0.298	0.360	0.293
Image only: ViT-Base/16	768	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577	0.733	0.615	0.667	0.577

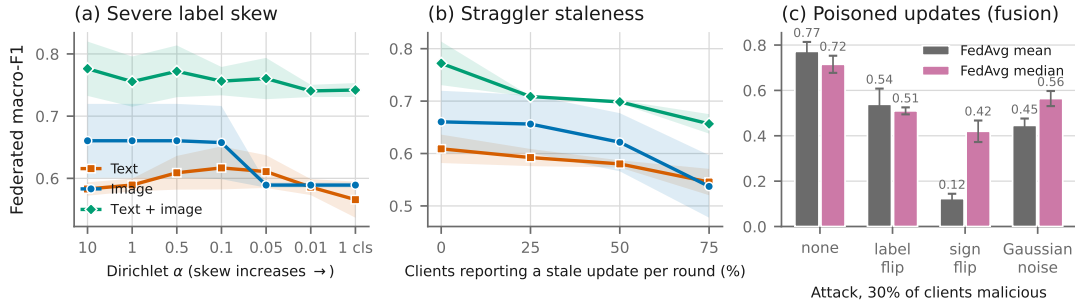


Fig. 17: Federated failure modes on the audited split. (a) Label skew from $\alpha=10$ to one class per client. (b) Clients reporting an update computed from an earlier global model. (c) Malicious clients under mean against coordinate-wise-median aggregation. Bands are ± 1 SD over 3 seeds. Client dropout is omitted because it is flat to 70%; the released figure set carries it.

TABLE VI: Validation-only family selection; test follows selection. Sparse-note retention is 15%. The image baseline is ViT-tiny, unlike ViT-Base in Table IV.

Actual system	Val. acc.	Test acc.	Test F1
Text PLM (DistilBERT)	0.2297	0.2400	0.2167
Image ViT (ViT-tiny)	0.7432	0.6400	0.5532
Raw ResNet+compact VLM	0.7568	0.5867	0.5076
ViT+DistilBERT VLM	0.7297	0.6400	0.5334
Routed ResNet+compact (LEAF)	0.7703	0.6133	0.5319

against class-level 0.600/0.613: the embedding is organized by class, not by crop identity.

The mismatch result shows dependence on the note,

not whether the image is used at all. We evaluate the retained checkpoint on the locked 75-crop test under graded corruption of each modality and two full image ablations, comparing predictions crop-by-crop against the clean run.

Both branches are load-bearing. Table VII reports the sweep. Deleting every attended text token costs 0.226 macro F1 and moves 41 of 75 predictions; replacing the image with uniform noise costs 0.429 and moves 61. The image now carries more of the decision than the note, reversing the previous corpus, but neither branch is inert. The reliability gate does not track this: mean w_t is 0.263 on

TABLE VII: Retained-checkpoint corruption on 75 locked test crops. Flips count changes from clean predictions. Full removal affects both modalities more strongly than partial corruption.

Condition	Acc.	Macro F_1	Flips
Clean	0.6267	0.5163	—
Text deletion 25%	0.6133	0.5060	4/75
Text deletion 50%	0.6267	0.5244	7/75
Text deletion 75%	0.6000	0.5068	11/75
Text deletion 100%	0.3600	0.2903	41/75
Image Gaussian $\sigma=0.09$	0.6400	0.5234	13/75
Image Gaussian $\sigma=0.18$	0.6400	0.5308	16/75
Image Gaussian $\sigma=0.26$	0.6267	0.5247	17/75
Image Gaussian $\sigma=0.35$	0.5867	0.4906	25/75
Image zeroed	0.3733	0.1603	49/75
Image uniform noise	0.2800	0.0875	61/75

clean input and rises to 0.361 under complete text deletion and 0.380 under complete image noise, so it weights a destroyed modality more rather than less.

Ablation strength determines the verdict. Additive Gaussian noise at $\sigma=0.35$ leaves leaf and lesion plainly visible and moves 25 of 75 predictions, whereas zeroing the image moves 49 and replacing it with uniform noise moves 61. An earlier 30-pair proxy screen used the Gaussian setting alone and read the visual branch as inert; that reading does not survive genuine removal. Evidence that a branch is unused requires an ablation that removes it, not one that degrades it.

I. A Third Modality That Carries No Signal

The reliability router is meant to express how much each modality is worth on a given example. Sections III-B and the missing-modality probes test it with two informative inputs; this subsection tests the complementary case, where a modality is known in advance to be worthless, so the correct behaviour is known by construction.

Construction. We attach daily observations from the AMFU Kharagpur station (IMD 42893), the observatory nearest the garden, as a third input: a 19-dimensional standardized vector of temperature, humidity, rainfall and its 7- and 30-day accumulations, rainless-spell length, vapour pressure deficit, evaporation, soil temperature and sunshine hours. A small multilayer perceptron embeds it to the fusion width, the router widens to $r_t + r_v + r_w = 1$, and the fusion head gains one block. Because the corpus records no capture date (Section III-A), no observation can honestly be joined to a photograph. Each crop is therefore assigned a station day by hashing its (photograph, box index) pair — independently of its label. We verify the independence rather than assert it: per-class mean weather agrees to within 0.28 °C and 0.43 h of sunshine, and the share of crops whose assigned day is high-risk for their own class is 12.1% against a 12.0% base rate. The modality is noise with respect to the target, by construction and by measurement.

Accuracy is unchanged, and seed spread dominates. Table VIII reports five seeds per arm. Every difference is at most a tenth of the pooled standard deviation: +0.011 on validation, +0.005 on test, +0.003 federated. The standard deviation across seeds (0.07–0.08 macro F_1) is an order of magnitude larger than any gap between the arms, and the sign of the difference changes from seed to seed (Figure 18a). A single-seed comparison on this corpus can therefore report a gain or a loss of several points from

TABLE VIII: Weather-modality ablation on the tea corpus, mean $\pm$ standard deviation over five seeds. The crop-to-station-day assignment is synthetic and label-independent, so weather carries no information about the target; the arms are statistically indistinguishable and the seed spread exceeds every difference between them.

Metric	Text+image	Text+image+weather
Validation F_1 (macro)	0.693 $\pm$ 0.079	0.703 $\pm$ 0.066
Federated F_1 (macro)	0.573 $\pm$ 0.070	0.576 $\pm$ 0.097
Test F_1 (macro)	0.570 $\pm$ 0.080	0.575 $\pm$ 0.034

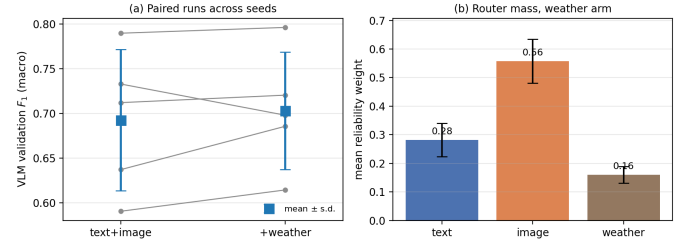


Fig. 18: Weather-modality probe over five seeds. (a) Paired runs; grey lines join the two arms for one seed, the square marks the mean with one standard deviation. The arms overlap and the sign of the difference flips across seeds. (b) Mean reliability mass per modality in the weather arm: the router gives 0.16 to an input that is provably uninformative.

noise alone, which is a caution that applies to the small-corpus tables elsewhere in this paper as much as to this one.

The router does not reject it. The informative result is Figure 18b. Averaged over seeds the gate assigns 0.16 ± 0.03 of its mass to weather, rather than the zero that an ideal reliability estimate would give a modality containing nothing. The gate is trained only through the classification objective, so nothing pushes an uninformative input’s weight to zero as long as the fusion head can compensate; the learned weights are a statement about optimization, not a calibrated measure of evidence. They should therefore be read as diagnostics to inspect rather than as trustworthy per-modality reliabilities.

J. Architecture Component Ablation

Eleven variants $\times$ 3 seeds on one device (33 runs) separate component effects from seed noise, which the earlier CPU-versus-H100 comparison confounded. The floor is high: pooled within-variant SD is 0.032 accuracy and the full model alone spans 0.053 across seeds (0.649 mean). One effect clears it — a lightweight vision encoder costs 0.227 — and no other ablation moves the mean by more than 0.071. The expert-residual weight is unsupported: $\lambda=0$ (−0.013) beats $\lambda=1$ (−0.044) and $\lambda=4$ (−0.040) against the $\lambda=2$ default, so the ordering is non-monotone. With a frozen-encoder concatenation reaching 0.800 on the same support (Table V), the trained architecture is not what produces the multimodal result and is not claimed to be.

K. Advisory Retrieval Results

Retrieval reaches 90.05% correct-class precision@1 against a 20.01% chance rate, rising to 95.52% with the classify stage. That second number is the classifier’s accuracy, because retrieval restricted to one predicted class can only return that class, so routing rather than

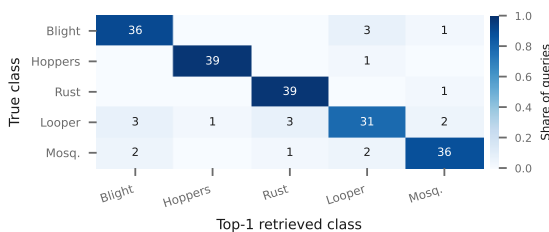


Fig. 19: Advisory retrieval confusion: true class against top-1 retrieved class, row-normalized so the diagonal is per-class precision@1; cells are query counts. Looper caterpillars is the weak row (31/40); hoppers and rust retrieve almost perfectly.

ranking decides correctness. A stratified query audit locates the failure: open retrieval is near-saturated (99.24%) on the queries carrying a class-unique sentence and falls to 72.46% on the rest, which routing lifts to 89.86%. Figure 19 shows where the misses go: looper caterpillars is the weak row, losing queries to blight, rust and mosquito bug, whose damage vocabulary it shares.

Three limits apply. The corpus is template-composed, so base and queries share sentence vocabulary even though no observation is shared, and these are upper bounds relative to free-form notes. The fusion re-ranker, the OBB overlay and the agronomic correctness of the advice remain unevaluated: this is a retrieval result, not evidence that the treatment is right.

L. Reproducibility and Threats to Validity

The released bundle records data identity, checkpoint state, every federated round and each realized client partition; deterministic grouping, strict reload and validation-only adaptation are reproducible, but external estate transfer is not inferred from them. The locked test is small (38 sources, 75 crops, two leaf-hopper crops), and selection on 74 validation crops can overfit. The central checkpoint has one seed; FL seeds vary only client partitions, and clients are simulated. DistilBERT is a PLM rather than a generative LLM. The corruption probe is one seeded realization per severity and carries no interval. The robustness sweep federates linear probes on frozen features, so its invariances belong to that estimator and not to the deep model. Deployment requires multi-estate data, independent notes, rare-class coverage, and repeated end-to-end training.

VII. Conclusion

On 371 linked crops, validation retains LEAF, the ResNet-compact-Transformer router; the ViT-DistilBERT candidate’s 2-crop test edge is non-significant ($p = 0.82$), and only vision-encoder capacity clears the 33-run seed-noise floor. FedAvg retains 84–94% of the 0.6919 initialization over 2–8 simulated clients. Fused federated training absorbs severe label skew and 70% client dropout (0.741–0.776 macro F1) but not staleness or poisoning: 30% sign-flipping clients cut it from 0.772 to 0.123, and coordinate-wise median recovers only part of that. In the matched suite FedAvg averages 0.510/0.492 at $\alpha = 0.1/1$, warm start no longer helps, and updates cost 18.9–67.2 MiB; no E3 arm collapses, so the stack is not claimed necessary. FarmFederate demonstrates a mobile path from photo and symptom capture to results

and advisory follow-up, with simulated monitoring views kept separate. Multi-estate, co-observed evaluation and a measured farmer-facing deployment remain required.

References

- [1] J. Yang, G. Xu, M. Yang, and Z. Lin, “Lightweight wavelet-CNN tea leaf disease detection,” *PLOS ONE*, vol. 20, no. 5, art. e0323322, 2025.
- [2] Y. Hu, J. Tang, and J. Yang, “Introducing artificial intelligence technology to plant disease management for sustainable agriculture,” *Crop Protection*, vol. 184, art. 106764, 2024.
- [3] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [4] H. A. Tahir, W. Alayed, and W. U. Hassan, “A federated explainable AI framework for smart agriculture,” *IEEE Access*, vol. 13, pp. 97567–97584, 2025.
- [5] H. Devaraj et al., “RuralAI in tomato farming: Integrated sensor system, distributed computing, and hierarchical federated learning,” *IEEE Sensors Lett.*, vol. 8, no. 5, pp. 1–4, 2024.
- [6] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. AISTATS*, 2017, pp. 1273–1282.
- [7] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [8] A. Radford et al., “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021, pp. 8748–8763.
- [9] J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Proc. ICML*, 2023.
- [10] Y. Wang et al., “A large language model for multimodal identification of crop diseases and pests,” *Scientific Reports*, vol. 15, art. 21959, 2025.
- [11] L. Li, J. Li, D. Chen, L. Pu, H. Yao, and Y. Huang, “FedReplay: A feature replay assisted federated transfer learning framework for smart agriculture,” *arXiv:2511.00269*, 2025.
- [12] H. Jégou, “Training data-efficient image transformers & distillation through attention,” in *Proc. ICML*, 2021, pp. 10347–10357.
- [13] Z. Liu et al., “Swin Transformer: Hierarchical vision transformer using shifted windows,” in *Proc. ICCV*, 2021, pp. 10012–10022.
- [14] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” in *Proc. CVPR*, 2022, pp. 11976–11986.
- [15] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. ICML*, 2019, pp. 6105–6114.
- [16] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT,” *arXiv:1910.01108*, 2019.
- [17] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. EMNLP*, 2019, pp. 3982–3992.
- [18] P. Garg, A. Mishra, R. Raja, A. Kumar, M. V. Joshi, and V. S. Palaparthi, “Multimodal data fusion by integrating IoT-enabled sensors and images for Jamun crop disease detection with machine learning,” *IEEE Trans. AgriFood Electron.*, vol. 3, no. 2, pp. 582–590, 2025, doi: 10.1109/TAFE.2025.3585065.
- [19] A. K. Pandey, G. D. Sinniah, A. Babu, and A. Tanti, “How the global tea industry copes with fungal diseases—Challenges and opportunities,” *Plant Disease*, vol. 105, no. 7, pp. 1868–1879, 2021.
- [20] R. Dembani, I. Karvelas, N. A. Akbar, S. Rizou, D. Tegolo, and S. Fountas, “Agricultural data privacy and federated learning: A review of challenges and opportunities,” *Comput. Electron. Agric.*, vol. 232, art. 110048, 2025.